

YONG-HWA PARK, Ph. D

Tenured Associate Professor
Human-Machine Interaction (HUMAN) LAB
Director of Center for Noise and Vibration Control Plus (NOVIC+)
Director of LIG Defense and Aerospace – KAIST Research Center (LIG D&A-KAIST)
Department of Mechanical Engineering
Korea Advanced Institute of Science and Technology (KAIST)
291 Daehak-ro, Yuseong-gu, Daejeon 34141, Korea
Office: +82-42-350-3235, Mobile: +82-10-3326-7337
Email: yhpark@kaist.ac.kr, Homepage: human.kaist.ac.kr

BIOGRAPHY

Yong-Hwa Park is Tenured Associate Professor of the Department of Mechanical Engineering at Korea Advanced Institute of Science and Technology (KAIST), and directors of the Center for Noise and Vibration Control Plus (NOVIC+) and the LIG Defense and Aerospace Research Center. He is also an adjunct professor with the Department of AI, the Robotics Program; the Graduate School of Advanced Security Science and Technology; and the Industrial Liaison Program (ILP) at KAIST. He serves as a vice-president of the Korean Society for Noise and Vibration Engineering (KSNVE); a vice-president of the Dynamics, Control and Robot Division of the Korean Society of Mechanical Engineers (KSME). He is an associate editor of JSMT (Springer), JMST-Advances, and KSME Journal. He has been serving as a conference chair of SPIE Photonics West since 2013, and served as general co-chair of International Congress on Sound and Vibration in 2025 (ICSV31). He is appointed as a program manager of Applied Mechanics Division at the National Research Foundation of Korea (NRF). in 2022, He was awarded by the *Minister of Science and ICT of the Korean government for his outstanding contributions to Korean science and engineering.*

Yong-Hwa Park holds BS, MS, and Ph. D from KAIST in 1991, 1993, and 1999, respectively, majoring in theoretical and experimental dynamical systems. During 2000-2003, he worked as a research associate at the Department of Aerospace Engineering Sciences, University of Colorado at Boulder, carrying out research on reduced-order modeling of dynamical systems and high-fidelity simulation for the optical microsystems. in 2003, he joined the Visual Display Division of the Samsung Electronics R&D Headquarter as a project leader of the micro-opto-electro-mechanical systems (MOEMS) for the development of Digital Light Processing (DLP) TV, the first and the best such technology in the world. From 2005 to 2016, Dr. Park carried out several innovative research thrusts in a capacity of as a *Research Master at the Samsung Advanced Institute of Technology (SAIT)*, achieving more than 70 international patents and several product commercialization of Lidar, 3D imaging, and wearable health care devices.

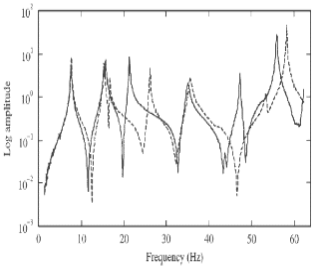
Since joining KAIST in 2016, He has been concentrating his teaching and research focused on vibrations, acoustics, and optical measurements, infusing his industrial innovation experience into his teaching and academic research relevance. Specifically, his research goal has been to understand the dynamical behavior such as vibration and sound from human and machines to predict their conditions with high-fidelity. The strategy is a bottom-up integration from the fundamentals of mechanics and measurement technology to the inference of conditions by the state-of-the-art deep learning. This forms his focused research topics of *high-fidelity vibration analysis; auditory intelligence; 3D optical measurement (Lidar); and condition monitoring applications in conjunction with the unique knowledge-based deep learning architecture.* His research accomplishments to date include: publications of *55(25) SCIE(domestic) Journal papers; 119(202) international(domestic) conference proceedings including 2 keynotes, 16(2) invited papers, 6(19) best paper awards; and 159 patents worldwide.* From his research works, he accomplished about *109 R&D projects with research grants of total USD 20.62M (11.23M as PI)* mostly from National Research Foundation (NRF), Agency for Defense Development (ADD), Hyundai, Samsung (honored him the best project award in 2023), LG Electronics, Inbody Inc., and others.

in teaching, since joining KAIST, he has developed several innovative new curricula that integrate classical mechanics (mechanical vibration, random data processing), with industrial innovations (main instructor of capstone design, AI in mechanical engineering, lecture series on semiconductor photonics such as Lidar, condition monitoring), and interactive discussions in virtual space (so called Education 4.0Q). He was awarded as the best lecturer in 2021 and 2023 spring semesters in Department of Mechanical Engineering, KAIST. *He has been supervising 28 doctoral students and 43 MS students since 2016*; the graduates among them are employed at Samsung Electronics (11), LG (2), Hyundai (3), Startups (4), major research institutes in Korea (2), and university faculties (2).

RESEARCH THEME

The Human-Machine Interaction (HUMAN) lab's research goal is to model, analysis, and acquire the dynamics of systems with high-fidelity simulation and measurements to identify the conditions of machines and human body through vibration, sound, and optical means. The focused research topics include *high-fidelity vibration analysis*, *auditory intelligence*, *health monitoring* and *3D optical measurement (Lidar)* with applications to *deep learning-based condition monitoring*. Machine condition monitoring and fault diagnosis are strongly pursued through the vibration and acoustic domain-knowledge in conjunction with physics-informed deep learning. We consider the human body as a complex dynamical system to be naturally interested in the analysis of human body's sound and vibration to predict the health conditions in a non-invasive way. As a novel measurement technology, 3D optical measurement (Lidar) is developed for the 3D remote sensing applications. The state-of-the-art physics-informed deep learning architectures are intensively developed as a platform algorithm for the inference of the conditions. All research topics are based on the strong foundations of vibration, acoustics and optical measurement aiming for the attractive industrial applications under interdisciplinary collaborations across 4 research groups in the Human lab as follows:

● High-fidelity Vibration Analysis and Vibration-based Condition Monitoring



As theoretical foundation, vibration analysis of the structures focusing on inverse problem is studied. High-fidelity finite element simulation framework of multi-dimensional problems (time-frequency-parameter dimensions) is developed using a novel domain decomposition (PGD). Experimental modal analysis, finite element model updating, and parameter identification methods are studied as key building blocks. These provide a unique computational framework so-called high-fidelity digital twin for condition monitoring and prognostics and health management (PHM) in conjunction with the unique vibration knowledge-based deep learning architecture.

● Auditory Intelligence and Sound-based Event Detection



Targeting human-like robust sound event detection against harsh listening conditions in real world, we utilize knowledge of the human auditory systems for sound classification, localization, and separation in conjunction with the state-of-the-art neural network in time-frequency-spatial domain. We pursue the binaural (2-ears) sound recognition for humanoid robot applications by in-depth functional analysis of human auditory system including head related transfer function (HRTF), various auditory filters and auditory cortex neural responses. Machine diagnosis, person/risk identification, health sensors, humanoid robot and metaverse are relevant project applications.

● Human Sound and Vibration for Health Monitoring



Non-invasive health monitoring methodology by means of human sound and vibration is studied including cardiovascular health monitoring (blood pressure, vascular age), continuous glucose monitoring, and respiratory diseases detection in Covid-19. Intensive coupled-field simulation, bio mimicking hardware prototyping (virtual patient) and time-frequency signal analysis are articulated in an interdiciplinary fashion. To effectively model the nonlinear vibration of the human body, the deep-operational neural net (Deep-O-Net)-based architecture is been developed. Wearable, non-invasive, continuous cardio-vascular health monitoring devices are expected outcomes

● 3D Optical Measurement (Lidar) and Remote Sensing



As the Human lab's well-established measurement technology, a unique 3D Lidar architecture is developed as key enabling technology for the remote sensing of human and machine behaviors. The target is miniaturization of the optical system tackling various crucial problems in Lidar products, by providing compact, low-cost, with high-resolution utilizing patented novel optical system and parallel-phase demodulation architecture. Number of projects have been executed as high-precision 3D measurement for autonomous vehicles; robots; manufacturing process monitoring; machine diagnosis; and health monitoring.

EDUCATION

3/1993–8/1999 ***Ph. D* in Mechanical Engineering, KAIST**

Dissertation: *Optimal Structural Dynamics Modification Using Frequency Response Function Synthesis and Its Applications*, supervised by Prof. Youn-sik Park. Conducted Finite element method-based design sensitivity analysis, experimental modal analysis, structural optimization and parameter identification to suggest FRF-based optimal structural design and identification.

3/1991–2/1993 ***M. S.* in Mechanical Engineering, KAIST**

Thesis: *A Study on the String and Beam Vibration Analysis subject to Moving Boundaries Using Wave Propagation Method*, supervised by Prof. Youn-sik Park. Accomplished transient vibration analysis of deploying structural members. A hybrid analytic-numerical approach is suggested to consider the time-varying boundary conditions of vibrating structures.

3/1987–2/1991 ***B. S.* in Mechanical Engineering, KAIST**

Qualifying thesis: *Damage Detection Using Gauss Quadrature*, supervised by Prof. Yang-Han Kim. Conducted vibration signal-based structural health monitoring. Damage location and severity are identified by Gaussian filters on the measured vibration signals.

CAREER PATH

8/2016–present ***Tenured Associate Professor*** (2024-), Department of Mechanical Engineering, KAIST. Leading the Human-Machine Interaction (HUMAN) LAB.

1/2023–present ***Director***, Center for Noise and Vibration Control (NOVIC+), ME, KAIST.

3/2026–present ***Director***, LIG Defense and Aerospace – KAIST Research Center, KAIST.

4/2026–present ***Adjunct Professor***, Department of AI, KAIST.

12/2023–present ***Adjunct Professor***, Graduate School of Advanced Security Science and Technology, KAIST.

12/2021–present ***Adjunct Professor***, Robotics Program, KAIST.

9/2017–8/2021 ***Adjunct Professor***, Institute for Artificial Intelligent, KAIST.

2/2012–8/2016 ***Research Master***, Samsung Advanced Institute of Technology (SAIT), Suwon, Korea. Research and developments of future 3D imaging devices and systems. Conducted 3 commercialization projects utilizing 3D image sensor.

3/2005– 1/2012 ***Principal Engineer***, SAIT, Yongin, Korea. Developments of optical resonators for 3D image sensor and resonating MEMS scanner for laser display systems.

7/2003 –2/2005 ***Senior Engineer***, Visual Display Division, Samsung Electronics, Suwon, Korea. Development of micro-display devices and next generation Digital Light Processing (DLP) systems with LED/LD light sources. Contributed to launching the world-first LED-based DLP TV. Research and business development of new display products.

9/2000 –6/2003 ***Postdoctoral Research Associate***, Center for Aerospace Structures, Department of Aerospace Engineering Sciences in University of Colorado at Boulder, supervised by Prof. K. C. Park. Conducted structural dynamics optimization and parallel computing. Coupled field structural dynamic analysis and design of optical microsystems. Conducted group research with CAMPmode (Center for Advanced Manufacturing and Packaging of Microwave, Optical, and Digital Electronics for Microstructures) with Prof. K. C. Park, Victor Bright and Prof. Y. C. Lee. Optical & RF resonating MEMS design and multi-disciplinary micro-systems simulation.

9/1999 –8/2000 ***Postdoctoral Research Associate*** *(during military service)*, Acoustics and Vibration Laboratory, Korea Research Institute of Standard and Science (KRISS). Human body vibration analysis and experiments, parameter identification, and standardization of human vibration.

3/1991 –8/1999 ***Research Assistances,*** Center for Noise and Vibration Control (NOVIC), ME, KAIST, Taejon, Korea. Conducted government projects (including G7-project) for design and modal analysis of next-generation low noise and vibration vehicles.

PROFESSIONAL ACTIVITIES

KAIST

Program Director	Industrial Liaison Program (ILP), KAIST. 2021.9-present.
Chairman	Industry-Academia Collaboration Board of ME KAIST, 2017.7-2025.8
Chairman	Cafeteria Improvement Committee of KAIST, 2024.4-present
Core Professor	KAIST Institute for AI, 2019.01-2021.9
Director	Advanced Video Industry Technology Research Center, KAIST, 2018.9-2023.12
Member	Executive Committee of the KAIST Convergence Research Center, 2023.9-present
Member	Establishment Committee of the AI Graduate School, 2018.6-2020.6
Member	Establishment Committee of the KAIST AI Research Hub, 2020.8-2020.12
Member	Establishment Committee of Manufact. Big Data AI Supercomputer Center, 2019.8-2020.7

Academia

General co-chair	International Congress on Sound and Vibration(ICSV31), 2025.7
Conference Chair	SPIE Photonics West, MOEMS and Miniaturized Systems, 2013-present
Senior Vice-President	Korean Society of Mechanical Engineers(KSME), Dynamics, Control & Robot Div. 2016-.
Vice-President	Korean Society for Noise and Vibration Engineering (KSNVE), 2017-present
Org. Comm. Chair	Korean Society for Noise and Vibration Engineering (KSNVE), 2023-2025
Theme Organizer	International Congress on Acoustics (ICA), 2022-present
Local Org. Comm.	International Congress on Sound and Vibration (ICSV 2025), 2023.1-2025.7
Int. Steering Comm.	Asia-Pacific Vibration Conference (APVC), 2019-present
Local Org. Chair	International Conference of Motion and Vibration Control (MOVIC), 2017-2018
Program Committee	International Meeting on Information Display (IMID), 2006
Series Editor	KAIST Research Series, (Springer-Nature), 2023-present
Associate Editor	Journal of Mechanical Science and Technology (JMST, Springer), 2023-present
Associate Editor	JMST Advances (Springer), 2018-present
Associate Editor	KSME Journal, Dynamics Division, 2017-present.
Editorial Board	Int. Journal of Precision Engineering and Manufacturing (IJPEM, Springer), 2017-2018
Board Member	KSME, Biological Engineering Division, 2023-present.
Board Member	KSME, Artificial Intelligence Research Association, 2020-present
Board Member	Korean Society of Creative Application, 2017-present

Government

Program Manager (PM)	National Research Foundation of Korea (NRF), Engineering Division, Applied Mechanics Field, Research Board (RB), 2023. 11. 1-present
Council Member	Korean Army Headquarter, Science and Technology Development Council, 2023.2-present
Committee Member	Daejeon City-KAIST Strategic Business Research Center, Executive Committee, 2023.9-.
Committee Member	Korean Urban Air Mobility (K-UAM) Core Technology Development, Project Planning Committee, 2021.01-2021.12

AWARDS (56 including Government Ministerial Commendation)

Ministerial Commendation	**Ministry of Science and ICT of Korean Government**, Ministerial Commendation for the Outstanding Contribution to Advancement of the Artificial Intelligence Industry, 2022.12.13
Best Lecturer Award	**ME, KAIST**, ME351 Mechanical Vibrations, 2023.10.31
Best Lecturer Award	**ME, KAIST**, ME351 Mechanical Vibrations, 2021.08.26
Best-Teacher Award	**KAIST,** The 53rd anniversary of the university's founding ***Lim Hyungkyu LINKGENESIS Best-Teacher Award.*** 2024.02.14

Globalization Excellence Lab	**KAIST**, The 52rd anniversary of the university's founding, 2024.02.14
Best PhD Dissertation Award	**College of Engineering, KAIST**,2024.02.21
Outstanding Doctoral Thesis Award	**KSME (Vision),** 2026 Spring KSME Conf., Samcheok, Korea 2026.4.23
Outstanding Doctoral Thesis Award	**KSME (Dynamics),** 2024 Spring KSME Conf., Yeosu, Korea, 2024.2
Outstanding Doctoral Thesis Award	**KSME (Dynamics),** 2023 Spring KSME Conf., Busan, Korea, 2023.4.27
Honored Best Paper Award	**Korean Society for Noise and Vibration Engineering (KSNVE**) in field of Vibration & Control, Korea, 2025.11.27
Honored International Academic Award	**KSNVE**, Korea, 2025.11.27
Best Contribution Award	**KSNVE**, 2026 Spring Conference, Samcheok Korea, May 28, 2026
Best Contribution Award	**KSNVE**, 2022 Spring Conference, Changwon Korea, May 26, 2022
Best Paper Award	**SPIE Photonics West**, MOEMS and Miniaturized Systems, San Francisco, 2012.1.25
Best Paper Award	**ICSV** (Bio), The 31st ICSV Conf., Incheon, Korea, 2025.11
Best Paper Award	**ICSV** (Measurement), The 31st ICSV Conf., Incheon, Korea, 2025.11
Best Paper Award	**ETME** (Bio), 2025 ETME Conf., Jeongseon, Korea, 2025.12.10
Best Student Paper Award	**SPIE Photonics West**, MOEMS and Miniaturized Systems, San Francisco, 2023.1.31
Best Paper Award	**KSME** (Bio), 2025 Fall Conf., Jeongseon, Korea, 2026.04.23
Best Paper Award	**KSME** (Dynamics), 2025 Spring Conf., Jeju, Korea, 2026.04.23
Best Paper Award	**SNAK,** The Society of Naval Architects of Korea, 2024.11
Best Paper Award	**ACME,** The 11th Asian Conference on Multibody Dynamics, 2024.08
Best Paper Award	**KSNVE**, KTM Tech. Best Student Presentation Award, 2023.10.12
Best Paper Award	**KSNVE**, KTM Tech. Best Student Presentation Award, 2022.11.03
Best Paper Award	**KSME** (IT), 2023 Spring Conference, Busan, Korea, 2023.4.27
Best Paper Award	**KSME** (Bio), 2022 Fall Conference, Jeju, Korea, 2022.11.11
Best Paper Award	**KSME** (Bio), 2022 Fall Conference, Jeju, Korea, 2022.11.11
Best Paper Award	**KSME** (Dynamics), 2021 Fall Conf., Gwangju, Korea, 2021.11.3
Best Paper Award	**KSME** (Dynamics), 2021 Fall Conf., Gwangju, Korea, 2021.11.3
Best Paper Award	**KSME** (IT), 2021 Spring Conf., Busan, Korea, 2021.5.27
Best Paper Award	**KSME** (Dynamics), 2020 Fall Conf., Online, Korea, 2020.12.17
Best Paper Award	**ASK**, 2019 Fall Conference, 2020.11.6
Best Paper Award	**KSME**, 2019 Fall Conference, Jeju Korea, 2019.11.15
Best Paper Award	**ASK**, 2019 Fall Conference, Pyungchang, Korea, 2019.11.8
Best Paper Award	**ASK**, 2019 Fall Conference, Pyungchang, Korea, 2019.11.6
Best Paper Award	**The Soc. of Micro and Nano Syst.**, K-MEMS, Jeju Korea, 2013.4.4
Young Professionals Award	**IEEE Ind. Electronics Society**, Annual Conference 2025, Spain, 2025.10
Young Professionals Award	**IEEE Ind. Electronics Society**, Annual Conference 2022, Belgium, 2022.10.20
Young Scientist Award	**ICA** 2022, Gyeongju, Korea, 2022.10.24
Best Project Award	**ROND**, The 6th miliTECH challenge, 2025.12
Best Project Award	**ROND**, The 5th miliTECH challenge, 2024.12
Best Project Award	**Samsung Heavy Industry**, 2023 Project Awards Ceremony, 2023.4.25
Best Project Award	**KAIST Future Underground City Research Center**, 2022 Project Awards Ceremony, 2022.12.20.
Best Project Award	**KAIST Manufacturing Innovation Center**, Challenging Projects in

	Manufacturing Advanced and Intelligence, 2023.1.31
Grand Prize Award	**POSCO Holdings,** POSCO Holdings Industrial AI-Robot Competition, 2025.09
Grand Prize Award	**KAOSTS,** Future Marine Scientist Award, KAOSTS Joint Conference, 2025.05
Grand Prize Award	**KSME**, The First KSME-LG Future Home Tech Challenge, 2022.11.10
Bronze Award	**KSME**, The First KSME-LG Future Home Tech Challenge, 2022.11.10
Bronze Award	**KSME**, The Second KSME-LG Future Home Tech Challenge, 2023.11.2
Encouragement Award	**KSME**, The Second KSME-LG Future Home Tech Challenge, 2023.11.2
Bronze Award	**Ministry of Commerce Industry and Energy**, 2021 Excellent Industry-Academia Project of the Smart Yard Development Program, 2022.12.22
Innovation Patent Award (CTO)	**Samsung Electronics** (Annual Awards), 2014.1.5
Value Innovation Award (CEO)	**Samsung Electronics** (Annual Awards), 2011.11.28
Best Paper Award, Silver (CTO)	**Samsung Electronics** (Annual Tech. Conference), 2011.11.7
Innovation Patent Award (CTO)	**Samsung Electronics** (Annual Awards), 2010.1.5
Frontier Project Award (VP)	**Samsung Electronics** (Annual Awards), 2009.1.9

LECTURES

2025/Fall	Random Data: Analysis and Processing (ME591), ME KAIST
2025/Spring	ME Co-op X (INT482, INT495), ME KAIST
2025/Spring	Vibrations of Linear Systems (ME551), ME KAIST
2024/Fall	ME Co-op XI (INT482, INT495), ME KAIST
2024/Fall	Random Data: Analysis and Processing (ME591), ME KAIST
2024/Spring	ME Co-op X (INT482, INT495), ME KAIST
2024/Spring	Vibrations of Linear Systems (ME551), ME KAIST
2023/Fall	ME Co-op IX (INT482, INT495), ME KAIST – Domestic, International
2023/Fall	Random Data: Analysis and Processing (ME591), ME KAIST
2023/Spring	ME Co-op VIII (INT482, INT495), ME KAIST – Domestic, **International (Newly established)**
2023/Spring	Mechanical Vibrations (ME351), ME KAIST **(Best Lecture Award)**
2022/Fall	ME Co-op VII (INT482, INT495), ME KAIST – Domestic
2022/Fall	Random Data: Analysis and Processing (ME591), ME KAIST
2022/Spring	ME Co-op VI (INT482, INT495), ME KAIST – Domestic
2022/Spring	Mechanical Vibrations (ME351), ME KAIST
2021/Fall	ME Co-op V (INT482, INT495), ME, KAIST – Domestic
2021/Fall	Random Data: Analysis and Processing (ME591), ME KAIST
2021/Spring	ME Co-op IV (INT482, INT495), ME, KAIST – Domestic
2021/Spring	Mechanical Vibrations (ME351), ME, KAIST **(Best Lecture Award)**
2021/Spr-present	AI in ME II (Star-MOOC) **(Newly established)**
2021/Spr-present	AI in ME I (Star-MOOC) **(Newly established)**
2020/Fall	ME Co-op III (INT482, INT495), ME, KAIST – Domestic
2020/Fall	Random Data: Analysis and Processing (ME591), ME KAIST
2020/Spring	ME Co-op II (INT482, INT495), ME, KAIST – Domestic
2020/Spring	Artificial Intelligence in Mechanical Engineering (ME491), ME KAIST **(Newly established)**
2019/Fall	Capstone Design II (ME401, Professor in Charge), ME KAIST
2019/Fall	ME Co-op I (INT482, INT495), ME KAIST **(Newly established)**
2019/Fall	Semiconductor Photonics (PH616, Semiconductor Photonics, Joint Lectures)

2019/Spring	Vibration in Linear System (ME551), ME KAIST
2019/Spring	Capstone Design I (ME400, Professor in Charge), ME KAIST
2018/Fall	Renewable Ocean Energy System (ME597, Joint Lecture), ME KAIST
2018/Fall	Capstone Design II (ME401), ME KAIST
2018/Fall	Engineering Design (ME340, Professor in Charge), ME KAIST
2018/Spring	Capstone Design I (ME400), ME KAIST
2018/Spring	Mechanical Vibrations (ME351), ME KAIST
2017/Fall	Capstone Design II (ME401), ME KAIST
2017/Fall	Engineering Design (ME340, Professor in Charge), ME KAIST
2017/Spring	Capstone Design I (ME400), ME KAIST
2017/Spring	Mechanical Vibrations (ME351), ME KAIST
2016/Fall	Dynamics (ME251), ME KAIST
2017-2023	**6** Undergraduate Research Program (URP490) (including 2 Long-term URP), ME KAIST
2016-2023	**70** Undergraduate Individual Study (ME495), ME KAIST
2003/Spring	Computational Methods in Dynamics, Aerospace Eng. Sciences, Univ. of Colorado at Boulder, partial participation.

THESIS SUPERVISION (28 Ph. D, 43 Masters)

14 Doctoral theses completed, 14 Candidates:

Jeong Min Kim (2026.2)	*High-precision 3D Measurement Method for Large-Scale Structures Based on Stereo Vision (KAIST)*
Bomi Lee (2026.2)	*PPG Signal-based Vascular Health Monitoring Method and Its Application to Clinical Data Using Cardiovascular Simulator and Transfer Learning (KAIST)*
Daeguen Lim (2025.8)	*Proper Generalized Decomposition and DeepONet-based Generative Model for Condition Monitoring of Dynamical Systems (KAIST)*
Hyeonuk Nam (2025.8)	*Frequency Dynamic Convolutions for Sound Event Detection (Samsung Electronics)*
Byeong Yun Ko (2025.8)	*DNN-based HRTF Individualization Utilizing Pinna Images for Accurate Sound Source Localization (Samsung Electronics)*
Kang-Jae Park (2025.8)	*Sensor Placement and DOF Expansion Methods for the Realization of a High-Precision Digital Twin*
Hakjun Lee (2025.2)	*Road noise active noise control using road profile estimation*
Jae-Duck Lee (2024.8)	*Enhanced video motion magnification method for vision-based vibration measurement and its applications to modal analysis (Agency for Defense Development)*
Gyeong Tae Lee (2024.2)	*Binaural Sound Event Localization and Detection Neural Network based on HRTF Localization Cues for Humanoid Robots (Bujoen Electronics)*
Jae-Hak Jeong (2024.2)	*Fabrication of Cardiovascular Simulation for Reproducing Human Blood Pressure Waves, Constructing Blood Pressure Wave Database, and Inverse Estimation of Cardiovascular Parameters (Samsung Electronics)*
Seong-Hoo Kim (2024.2)	*Temporal Dynamic Convolutional Neural Network for Time Series Classification considering Non-Stationary Process (Samsung Electronics)*
Sung-Hyun Lee (2023.8)	*AMCW time-of-flight scanning LiDAR based on parallel-phase demodulation and its image enhancements (Samsung Electronics)*
Ik-Chae Jeong (2023.8)	*Performance measure for optimal placement of sensors and actuators of active noise control system (Samgsung Electronics)*
Gil-Yong Lee (2023.2)	*Proper generalized decomposition for multi-dimensional frequency response analysis (KAERI)*
Ph. D Candidates (**14**)	Wonjun Yi*(2020.9)*, Hyoun-Jun Kim*(2021.3)*, Junki Hong*(2022.3)*, Mintae Kim*(2023.3)*,

Yoon-Seop Lee*(2023.3)*, Wonho Jung*(2023.9)*, Changyu Yoon*(2024.3)*, Adelle Persad*(2024.9)*, Seok Jun Ahn*(2025.3)*, Jin Kim*(2025.3)*, Jun Ho Kim*(2025.3)*, Myounghee Kim*(2025.3)*, Hyunwoo Song*(2026.02)*, Jaeyoung An*(2026.02)*

30 Master theses completed, 13 Candidates:

Kangmin Jang (2026.2)	*PI-Deep Operator Network based Structural Modal Analysis for Various Structural Parameters (KAIST)*
Manhee Cho (2026.2)	*Generic calibration of wide-FOV camera-projector system for precision 3D measurement (KAIST)*
Hyunwoo Song (2026.2)	*Physics-Informed PPG-to-Central Blood Pressure Estimation Network Using Modular Blood Pressure Transmission Characteristics (KAIST)*
Seok Jun Ahn (2025.2)	*Ambient Light Noise Suppression in Depth Image of Indirect Time-of-Flight Sensors Using a Diffusion Model (KAIST)*
Jin Kim (2025.2)	*Image-based Sound Source Localization and Intensity Map Estimation Method Trained Using a Beamforming Array (KAIST)*
Jun Ho Kim (2025.2)	*Condition Monitoring of Intelligent Tires Utilizing Parallel 1D CNN and Vibration Measurement (KAIST)*
Adelle Persad (2024.8)	*Estimation of central blood pressure waveform using transfer function analysis of Photoplethysmography (ppg) signals at wrist (KAIST)*
Deokki Min (2024.2)	*Auditory Cortex Neural Response-Inspired Sound Event Detection*
Dohyeon Kim (2024.2)	*Stereo Matching Technique for Large-scale Ship Block Using Guided Mask in Keypoint-based ROI*
Changhee Min (2024.2)	*Deep Operator Network for Periodic Signal Problem with Application to Central Blood Pressure Monitoring*
Sung-Hyun Yoon (2024.2)	*Signal Imaging with Gramian Angular Field Based on the Current Signal and Envelopes for Anomaly Detection of Rotating Machine (Samsung Electronics)*
Nalin Kamath (2024.2)	*High-dimensional Bayesian Optimization for Finite Element Model Updating and Fault Detection (DTU-KAIST Dual Degree)*
Sung-Jin Cheong (2023.8)	*Infrared Image-based Remote Rescuee Detection Method in Maritime Situation Utilizing a Deep Learning Network and Data Augmentation (LG Innotech)*
Yoon-Seop Lim (2023.2)	*A Long-Distance 3D Face Recognition System Using a Novel High-Resolution Time-of-Flight Sensor (KAIST)*
Jae-Woong Bae (2022.8)	*Normal data-based ball bearing fault detection using CNN with self-labeling (Hyundai Motors)*
Junki Hong (2022.2)	*Estimation of Brachial Arterial Blood Pressure Waveform Using Transfer Function Analysis of Oscillometric Signals (KAIST)*
Sangmin Choi (2022.2)	*Time Data Masking Method using Interaural Cross-correlation for Sound Source Localization Algorithm using Head-related Transfer Function (Samsung Electronics)*
Wook-Hyeon Kwon (22.2)	*High Precision Time-of-Flight Sensor based on Multi-tap Parallel Demodulation (Samsung Electronics)*
Simeneh Gulelat (2021.2)	*Unsupervised Synthetic-to-Real Transfer Learning of Depth Images in Consideration of Sensor Characteristics (KAIST)*
Bomi Lee (2021.2)	*Correlation Analysis on Human Upper Limb Parameters and Oscillometric Signal During Blood Pressure Measurement (KAIST)*
Hyoun-Jun Kim (2021.2)	*Model Reduction of Fluid-Str.Coupled Syst. Using Dynamically Coupled Modal Basis (KAIST)*
Jeongmin Kim (2021.2)	*Correction of Image Distortion Using Virtual PCG Pattern in Stereo-based Large-scale Precision 3D Measurement (KAIST)*
Dae-Hee Park (2020.2)	*Removal of Reflected Virtual Images in Visual Recognition Utilizing 3D Depth Information (KAIST)*
Hye-Won Yoo (2020.2)	*Biometric Authentication utilizing Vibrational Characteristics of Human Fingers (Samsung Electronics)*
Hyeonuk Nam (2020.2)	*Quantitative Analysis of the Effect of Reverberation on Automatic Speech Recognition*

	Error Based on Coherence (KAIST)
In-Gyu Jang (2020.2)	*A Laser Scanning based 3D Object Sensing Method using Multiple Frequency Modulation (LG Innotech)*
Jun-Hyeok Lee (2020.2)	*Direction of arrival estimation utilizing deep neural networks with dual channel microphones under noise condition (Minds Lab)*
Seong-Hu Kim (2019.2)	*Speech separation from multi-speaker dialogues under reverberant environment (KAIST)*
Kang-Jae Park (2018.8)	*Measurement of Substance Concentration in Fluid using Structural System Parameter Identification: Application to Glucose Measurement (KAIST)*
Joon-Oh Shin (2018.8)	*A Methodology for Simultaneous Capturing of 3D Shape and Velocity Images of a Moving Object (Samsung Electronics)*
Master Candidates (**13**)	Yujeong Shin*(2024.3, KAIST-DTU Dual degree)*, Junhong Im*(2024.9)*, Inhan Choi*(2024.9, KAIST-DTU Dual degree)*, Yuri Song*(2025.3)*, Seungdeok Choi*(2025.3)*, Seongmin Choi*(2025.3)*, Minje Jin*(2025.3)*, Dongkyu Han*(2025.9)*, Sooyeon Jeong*(2025.9)*, Haein Shin*(2025.9)*, Jaemin Shin*(2026.3)*, Jonghyeok Park*(2026.3)*, Sejun Park*(2026.3)*

INVITED LECTURES · SEMINARS (111)

International (27):

07/2025	*High-Fidelity Digital Twin for Structural Health Monitoring Using Fusion of Vibrational Modal Analysis and AI* ***(Keynote),*** The 31st International Congress on Sound and Vibration (ICSV31).
04/2025	*High-fidelity surrogate model using vibrational modal analysis and AI for structural health monitoring* ***(Keynote),*** The 8th Korea-Japan Joint Symposium on Dynamics & Control.
09/2025	AI-based Condition Monitoring Research, 2025 HKU-KAIST Joint Student Workshop.
10/2024	*Vibration, Sound, 3D Vision-based Condition Monitoring*, National Taiwan Univ (NTU).
10/2024	*Vibration, Sound, 3D Vision-based Condition Monitoring*, Chang Gung University (CGU).
10/2024	*Vibration, Sound, 3D Vision-based Condition Monitoring*, Ming Chi University of Technology (MCUT).
02/2024	*Human Auditory System-Inspired Sound Event Detection and 3D Localization for Humanoid Robots*, Tohoku Uni.-KAIST Joint Workshop on 3D Audio and Binaural Technology Research.
12/2023	*Lidar for Autonomous Vehicles – Challenges and Innovations*, 2023 KAIST ME e-Symposium on Robotics & Autonomous Vehicles.
08/2023	*Partitioned Damage Identification of Structural Systems*, Joint Workshop on a New Displacement-based Partitioned (DP) Formalism and Related Research between Univ. of Colorado at Boulder-KAIST-SNU-KHU.
05/2023	*High Fidelity Digital Twin for Health Monitoring of Maritime Infrastructures*, 2023 Joint Scientific Seminar on Maritime Studies between KAIST and DTU.
01/2023	*Source Localization and Sound Event Detection for Humanoid Robots*, Tohoku Univ.
10/2022	*An accurate HRTF measurement for binaural source identification*, KAIST-DTU Joint Workshop for Acoustics, Hearing and AI.
10/2022	*Binaural source localization and event detection*, KAIST-DTU Joint Workshop for Acoustics, Hearing and AI.
10/2022	*HRTF individualization for metaverse applications*, KAIST-DTU Joint Workshop for Acoustics, Hearing and AI.
07/2022	*High Fidelity Digital Twin - Structural Vibration and System Identification for Fault Diagnosis*, Univ. of Colorado at Boulder
02/2022	*KAIST Research Overview for Jabra-KAIST Collaboration*, GN Audio, INC (Jabra, Denmark)

10/2020 *AI for Sound and Vibration: Cough Detection Camera (CNN); Person Identification using Finger Vibration (SVM)*, 2020 CAV workshop, Penn State Univ.

10/2020 *Noise and Vibration Researches at NOVIC+, KAIST*, 2020 CAV workshop, International Liaison Talks, Penn State Univ.

02/2019 *Acoustic Recognition: Smart Transducers and Artificial Intelligence*, KAIST-DTU Joint Workshop, ME KAIST

04/2018 *Coupled Field Vibration Analysis of Bio-systems for Non-invasive Measurement of Biomarkers*, Univ. of Colorado at Boulder, USA

11/2017 *Voice Recognition with Acoustic Metamaterial*, DTU, Denmark.

12/2016 *Machines Interacting with Human by 3D Recognition*, ITB, Indonesia

09/2018 *A Wearable Non-invasive Glucose Sensor Utilizing Coupled Field Vibration Analysis*, International Workshop on Micro/Nano-engineered Smart Sensors & Devices, KAIST

09/2018 *Speech separation in cocktail party under reverberant environment*, 2018 KAIST-KTH Joint Workshop on Vibro-acoustics, ME KAIST

09/2018 *Human-related Parameter Identification via Vibration Signal Processing for Health Sensor Applications*, 2018 KAIST-KTH Joint Workshop on Vibro-acoustics, ME KAIST

10/2017 *Human Machine Interaction*, KAIST-ITB Joint Workshop, ME KAIST

12/2016 *Machines Interacting with Human by 3D Recognition, ITB*

Domestic (84)

06/2026 *Robot-based Vision/Vibration/Acoustic Perception and Condition Monitoring Recent Technologies, KOMIPO-KAIST AX Competency Enhancement Program*.

10/2025 *Recent Advances in Multimodal Condition Diagnosis Based on Vibration, Acoustics, and Vision (POSCO Group Chair Professor Forum),* POSCO.

09/2025 *Generative Modeling of Dynamic Responses of Vibration Systems Using Physics-Informed DeepONet and Its Application to Condition Diagnosis (Invited Talk, PHM/Reliability Group),* KIMM.

08/2025 *Generative Modeling of Dynamic Responses of Vibration Systems Using DeepONet and Its Application to Condition Diagnosis (Invited Lecture),* Mechanical Artificial Intelligence Expert Symposium), KSME.

06/2025 *Development and Applications of Experiment-Coupled Simulation: Noncontact Ultrasonic Waveform Estimation and Vascular Health Assessment (SHDL Leading Research Center ERC Regular Seminar),* Hanyang University.

12/2024 *Auditory Intelligence and AI (Invited Talk, Voyager Institute for Convergence Medicine),* Hallym University Sacred Heart Hospital.

09/2024 *Development of Smart Condition Monitoring and Diagnostic Technologies (Advanced Seminar on Smart Ship and Autonomous Navigation Technology Development),* Samsung Heavy Industries.

07/2024 *Driver Health Condition Diagnosis Using Biometric Information (Focused on Automotive New Business Models) (Hyundai Motor Convergence Technology Expert Technical Advisory Seminar),* Hyundai Motor Company.

05/2024 *Human Auditory System-Inspired Sound Event Detection and Robotic Applications for Cognitive Science,* The 77th Annual Conference of the Korean Audiological Society.

04/2024 *3D LiDAR Technology,* The 34th Workshop on Optical Technology, Optical Society of Korea.

02/2024 *Ship Fault Diagnosis Technology (SHI-KAIST Tech Forum),* Samsung Heavy Industries.

01/2024 *Human Auditory System-Inspired Sound Event Detection and Localization for Humanoid Robot*

	Applications, KIST.
11/2023	*Advanced 3D Vision and Acoustic Sensors for Defense Advanced Intelligence*, Defense Agency for Technology and Quality.
10/2023	*Vibration, Acoustic 3D Information-Based Sensor and Recognition Technology*, Department of Mechanical Engineering, Hanyang University.
10/2023	*Vibration, Acoustic, 3D Information-Based Sensor and Recognition Technology*, Pusan National University (MERRIC).
03/2023	*High-Fidelity Digital Twin for Machine Health Monitoring*, MIDAS IT.
02/2023	*Vibration/Acoustic-based Weapon System Fault Diagnosis, Small/Precision LIDAR*, Army Headquarters
02/2023	*AI-Based Anomaly Detection and Applications Using Vibration, Acoustic, and Image Data*, LIG Nex1 Co., Ltd.
01/2023	*Autonomous Driving Lidar and Digital Health Technologies - 'Insights on CES 2023 Trends from Top Professors at KAIST'*, The Korea Economic Daily, Hankyung.
10/2022	*Advanced 3D Vision and Acoustic Sensors for Manufacturing Intelligence and Safety* (POSCO Distinguished Forum), POSCO CO., LTD.
08/2022	*Compressor Fault Diagnosis Technology*, LG Electronics Inc.
08/2022	*Binaural Source Localization and Sound Event Detection for Humanoid Robot* (The Korean Society for Noise and Vibration Engineering AI Summer Course 'Seeing Sounds'), The Korean Society for Noise and Vibration Engineering.
03/2022	*Motor Fault Diagnosis Technology*, LG Electronics Inc.
11/2021	*Advanced 3D Visual Sensor and Auditory Sensor* (Ministry of National Defense Senior Officials' Future Defense Strategy Innovation Program)), Ministry of National Defense/Eulji Research Center.
11/2021	*Human-Machine Interaction* (Small and Medium-sized Enterprise R&D Shared Center Technical Presentation 2nd), Four Major Science Institutes Joint - Small and Medium-sized Enterprise R&D Shared Center
09/2021	*Audio Recognition Technology*, ARE.
09/2021	*Human-Machine Interaction* (Small and Medium-sized Enterprise R&D Shared Center Technical Presentation 1st), Four Major Science Institutes Joint – Small and Medium-sized Enterprise R&D Shared Center.
09/2021	*Acoustic-based Fault Diagnosis Technology*, Samsung Heavy Industries Co., Ltd. (Ship and Ocean Research Institute).
08/2021	*Future Vibration and Acoustic-based Measurement, Diagnosis, and Recognition Technology* (Samsung Heavy Industries Technology Development Division Seminar), Samsung Heavy Industries Co., Ltd.
07/2021	*Vibration and Acoustic-based Fault Diagnosis Seminar*, Korea Institute of Industrial Technology (KITECH).
04/2021	*Artificial Intelligence Case Presentation: Human Sound and Vibration Recognition Technology*, Korean Society of Mechanical Engineers, winter school.
02/2021	*Human Sound and Vibration Recognition Technology*, Artificial Intelligence Research Society, Korean Society of Mechanical Engineers, winter school.
04/2020	*Advanced Technology and Industrial Trends Through CES 2020*, ME, KAIST
02/2019	*Advanced Technology and Industry Trends as Seen Through CES 2019*, Korea Institute of Machinery & Materials (KIMM)
2/2019	*CES 2019 Technology Trends*, KAIST Department of Mechanical Engineering Professor Workshop (AMEC)

11/2018	*Machines Interacting with Human by 3D Recognition*, Electronics and Telecommunications Research Institute (ETRI).
10/2018	*Speech Recognition Technology Simulating Human Auditory Perception and Future Research Challenges*, Seoul National University College of Medicine
08/2018	*Fundamentals of Vibration and Acoustics for Audiology*, Seoul National University College of Medicine
06/2018	*Machines Interacting with Human by 3D Recognition*, INTEKPLUS CO., LTD.
06/2018	*Machines Interacting with Human by 3D Recognition*, Korea Research Institute of Standards and Science (KRISS).
06/2017	*Human-Machine Interaction- Introduction to Future Technologies*, YUSHIN HIGH SCHOOL
11/2016	*About My Dream Regarding Robots*, Kind Science Research Institute.
11/2016	*3D Image Sensor for Human–Machine Interaction*, Korea Aerospace Research Institute (KARI)
08/2018	*Introduction to Future Technologies*, Science Gifted Camp Lecture, Global Institute for Talented Education, KAIST.
05/2017	*Machines Interacting with Human by 3D Recognition*, Invited Colloquium, CT, KAIST.
10/2016	*Putting Our Dreams on a Running Mouse, and About My Dream Regarding Robots*, MR KAIST
03/2019	*Trends in Advanced Technology and Industry Outlooks Through CES 2019*, Mechanical and Aerospace Invited Seminar, ME, KAIST.
07/2018	*Open lecture on mode analysis industry-academic cooperation - Basic Theory of Mode Interpretation and Computational mode analysis and dynamic structure change*, NOVIC KAIST.
03/2018	*Trends in Advanced Technology and Industry Outlooks through CES 2018*, Mechanical and Aerospace Invited Seminar, ME, KAIST.
06/2017	*Open lecture on mode analysis industry-academic cooperation - Basic Theory of Mode Interpretation and Computational mode analysis and dynamic structure change*, NOVIC KAIST.
04/2017	*Human Machine Interaction,* NOVIC KAIST.
10/2016	*Machines Interacting with Human: 3D Image Sensor*, NOVIC KAIST.
2/2012	*Recent Trends in Acoustics and Vibration Research*, KAIST Noise and Vibration Research Center Workshop and Others 2nd Session.
11/2011	*Micro Systems and Image Science*, College of Applied Science Kyung Hee University Convergence Workshop.
02/2011-10/2013	*Image Sensor Technology and Applications*, Samsung Advanced Technology Training Institute Expert Education Program, Total of 9 Sessions.
2008-2015	*Microsystems and Image Sensor Applications*, Business Commercialization Strategy, Targeting Various Business Units at Samsung Electronics, Over 20 Sessions.
05/2014	*KAIST Mentoring Concert*, KAIST Alumni Association, Corporate Sector Career Mentor.

INDUSTRY-ACADEMIC TRANING COURSES

2016.6-2025.7	The *34-41th Modal Analysis: Theory, Test Technology, and Application (Main instructor)* trained more than 300 industry engineers in the fields of vibration, acoustics; design and testing; measurement in Korean major manufacturing companies (Hyundai Motors, Samsung Electronics, Samsung Heavy Industry, LG Electronics, Hankook Tire Comp. and other Korean leading companies)

PUBLICATIONS

International Journals (55 SCI(E) Papers, 7 non-SCI, Total IF: 266.6), H-index 30 (Google Scholar)):

1. Junki Hong, Bomi Lee, Adelle Ria Persad, Hyunwoo Song, Changhee Min, Jae-Hak Jeong, Alireza Doostan, **Yong-Hwa Park**, "Robust prediction of parameterized cardiovascular hemodynamics using deep operator networks with time normalization", *Computer Methods and Programs in Biomedicine*, Vol. 280, 2026.06; **(SCIE, IF 6.4)**
2. K. C. Park, J. A. Gonzalez, S. J. Shin, S. H. Kang, M. H. Hwang, J. G. Kim, M. F. Baqir, Jae-Hung Han, Robel Weldebrhan Hagos, Seung Chan Lee, **Yong-Hwa Park**, HyeonJun Kim, K. K. Maute, C. A. Felippa, "Solving FEM models without assembly: Its promise and challenge", *Computer Methods in Applied Mechanics and Engineering*, Vol. 450, 2026.03; **(SCIE, IF 7.6)**
3. Jeongmin Kim, Jin Kim, Manhee Cho, Yeonghwan Kim, Bong-gook Kang, **Yong-Hwa Park**, "Three-dimensional measurement of ship blocks using a multi-vision system with weighted generic calibration", *Ocean Engineering*, Vol. 346, 2026.02; **(SCIE, IF 6.3)**
4. Wonjun Yi, Wonho Jung, Hyeonuk Nam, Kangmin Jang, **Yong-Hwa Park**, "Multi-output classification using a cross-talk architecture for compound fault diagnosis of motors in partially labeled condition", *Mechanical Systems and Signal Processing*, Vol. 244, 2026.01; **(SCIE, IF 10.2)**
5. Wonho Jung, Junho Kim, Kangmin Jang, Sung-Hyun Yun, Daeguen Lim, Minje Jin, **Yong-Hwa Park**, "Vibration, current, torque, RPM dataset for multiple fault conditions in industrial-scale electric motors under randomized speed and load variations", *Data in Brief*, Vol. 62, 2025.10; (**ESCI, IF 1.9**)
6. Hyeonuk Nam, Seong-Hu Kim, Deokki Min, Byeong-Yun Ko, **Yong-Hwa Park**, "Towards Understanding of Frequency Dependence on Sound Event Detection", *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, Vol. 33, pp. 3948-3960, 2025.09; **(SCIE, IF 5.2)**
7. Junki Hong, Bomi Lee, Adelle Ria Persad, Hyunwoo Song, Changhee Min, Jae-Hak Jeong, **Yong-Hwa Park**, "Correlation analysis of arterial pressure waveform morphology similarity and relative pulse transit time using cardiovascular hardware simulator and hemodynamic model", *Computers in Biology and Medicine*, Vol. 196, Part B, Article 110778, 2025.07; **(SCIE, IF 6.3 – Delisted Nov. 2025)**
8. Dae-Guen Lim, Gil-Yong Lee, **Yong-Hwa Park**, "Application of DeepONet to predict transient drop motion of the control rod in real-time", *Nuclear Engineering and Technology*, Vol. 57, No. 9, Article 103620, 2025.09; **(SCIE, IF 2.9)**
9. Sungjin Cheong, Wonho Jung, Yoon Seop Lim, **Yong-Hwa Park**, "Thermal-Infrared Remote-Target Detection System for Maritime Rescue Using 3-D Game-Based Data Augmentation With GAN", *IEEE Transactions on Geoscience and Remote Sensing*, Vol. 62, Article 5006113, 2024; **(SCIE, IF 8.6)**
10. Hyeonuk Nam, **Yong-Hwa Park**, "Coherence-based phonemic analysis on the effect of reverberation to practical automatic speech recognition", *Applied Acoustics*, Vol. 227, Article 110233, 2025.01; **(SCIE, IF 3.6)**
11. Byeong-Yun Ko, Deokki Min, Hyeonuk Nam, **Yong-Hwa Park**, "DNN-Based HRIRs Identification With a Continuously Rotating Speaker Array," *IEEE Transactions on Instrumentation and Measurement*, Vol. 74, 2025; **(SCIE, IF 6.1)**
12. K.C. Park, C.A. Felippa, S.H. Kang, J.A. Gonzalez, S.J. Shin, **Yong-Hwa Park**, J.G. Kim, "A partitioned

flexibility (PartFlex) method for structural analysis", *Computer Methods in Applied Mechanics and Engineering*, Vol. 429, Article 117155, 2024.09; **(SCIE, IF 7.6)**

13. Dae-Guen Lim, Gil-Yong Lee, Nam Gyu Park, **Yong-Hwa Park**, "A computational framework for drop time assessment of a control element assembly under fuel assembly deformations with fluid-structure interaction and frictional contact", *Nuclear Engineering and Technology*, Vol. 56, No. 8, pp. 3450-3462, 2024.08; **(SCIE, IF 2.9)**
14. Jeongmin Kim, Jaeduck Lee, **Yong-Hwa Park**, "Highly accurate three-dimensional measurement of large structures using multiple stereo vision with improved two-step calibration algorithm", *Measurement*, Vol. 234, Article 114886, 2024.07; **(SCIE, IF 6.1)**
15. Kangjae Park, Gil-Yong Lee, Jae Hak Jeong, **Yong-Hwa Park**, "Validation of the added mass effect in the narrow gaps of nuclear reactor pressure vessels", *Journal of Sound and Vibration*, Vol. 584, Article 118489, 2024.08; **(SCIE, IF 5.8)**
16. Wonho Jung, Sung-Hyun Yun, **Yong-Hwa Park**, "Vibration, and temperature run-to-failure dataset of ball bearing for prognostics", *DATA IN BRIEF*, Vol. 54, Article 110403, 2024.06; (ESCI, IF 1.4)
17. Gil-Yong Lee, Kang-Jae Park, Dae-Guen Lim, **Yong-Hwa Park**, "Model order reduction based on low-rank approximation for parameterized eigenvalue problems in structural dynamics", *Journal of Sound and Vibration*, Vol. 582, Article 118413, 2024.07; **(SCIE, IF 5.8)**
18. Kun Woo Ko, Taehoon Lee, Jae Hak Jeong, Yeonteak Lee, Ikjin Kim, Jungmok Seo, **Yong-Hwa Park**, Jin-Oh Kim, Steve Park, "Hydrogel Ink for 3D Printing with High and Widely Tunable Mechanical Properties via the Salting-Out Effect", *Advanced Materials Technologies*, Vol. 9, No. 10, 2024.05; **(SCIE, IF 6.0)**
19. Seong-Hu Kim, Hyeonuk Nam, Sang-Min Choi, **Yong-Hwa Park**, "Real-Time Sound Recognition System for Human Care Robot Considering Custom Sound Events", *IEEE Access*, Vol. 12, pp. 42279-42294, 2024; **(SCIE, IF 4.2)**
20. Jaeduck Lee, Kang-Jae Park, **Yong-Hwa Park**, "Enhanced video motion magnification utilizing envelop shift for vision-based vibration measurements", *IEEE Transactions on Instrumentation and Measurement*, Vol. 73, Article 5016814, 2024**; (SCIE, IF 6.1)**
21. Jae-Hak Jeong, Bomi Lee, Junki Hong, Changhee Min, Adelle Ria Persad, Tae-Heon Yang, **Yong-Hwa Park**, "Cardiovascular Hardware Simulator and Artificial Aorta-Generated Central Blood Pressure Waveform Database According to Various Vascular Ages for Cardiovascular Health Monitoring Applications", *Computers in Biology and Medicine*, Vol. 172, Article 108224, 2024.04; **(SCIE, IF 7.7 – Delisted Nov. 2025)**
22. Dae-Guen Lim, Gil-Yong Lee, **Yong-Hwa Park**, "Proper generalized decomposition-based iterative enrichment process combined with shooting method for steady-state forced response analysis of nonlinear dynamical systems", *Computational Mechanics*, Vol. 74, No. 5, pp. 937-953, 2024.11; **(SCIE, IF 4.5)**
23. Hyewon Yoo, Jae Woong Bae, Jae-Hak Jeong, **Yong-Hwa Park**, "Feasibility study on person identification utilizing frequency response functions of human fingers," *Journal of Sound and Vibration*, Vol. 568, 2024.01; **(SCIE, IF 5.8)**
24. D.G. Lee, K.C. Lee, J.W. Juhn, J.S. Lee, J. Kim, S. Park, M. Lehnen, **Yong-Hwa Park**, J.H, Jeong, J. Yim, Y.C. Ghim, "Development of the upgraded single crystal dispersion interferometer (SCDI-U) and its first measurements of the line integrated electron densities in KSTAR during shattered pellet injections," *Nuclear*

Fusion, Vol. 64, No. 1, 2024.1; **(SCIE, IF 4.1)**

25. Lee, Sung-Hyun, Yoon-Seop Lim, Wook-Hyeon Kwon, and **Yong-Hwa Park**, "Automatic Internal Stray Light Calibration of AMCW Coaxial Scanning LiDAR Using Black and White Image Patterns," *IEEE Transactions on Instrumentation and Measurement*, Vol. 73, 2023.12.18 **(SCIE, IF 7.0)**
26. K.C. Park, J.A. Gonzalez, **Yong-Hwa Park**, S.J. Shin, J.G. Kim, K.K. Maute, C. Farhat, C.A. Felippa, "Displacement-based partitioned equations of motion for structures: Formulation and proof-of-concept applications," *International Journal for Numerical Methods in Engineering*, Vol. 124, No. 22, 2023.11; **(SCIE, IF 3.3)**
27. Byeong-Yun Ko, Gyeong-Tae Lee, Hyeonuk Nam, **Yong-Hwa Park**, "PRTFNet: HRTF Individualization for Accurate Spectral Cues Using a Compact PRTF," *IEEE Access*, Vol. 11, 2023.8; **(SCIE, IF 4.2)**
28. Sung-Hyun Lee, Yoon-Seop Lim, Wook-Hyeon Kwon, **Yong-Hwa Park**, "Multipath Interference Suppression of Amplitude-Modulated Continuous Wave Coaxial-Scanning LiDAR Using Model-Based Synthetic Data Learning," *IEEE Sensors Journal*, Vol. 23, No. 19, 2023.8; **(SCIE, IF 4.3)**
29. Seong-Hu Kim, Hyeonuk Nam, **Yong-Hwa Park**, "Analysis-based Optimization of Temporal Dynamic Convolutional Neural Network for Text-Independent Speaker Verification," *IEEE Access*, Vol. 11, 2023.6; **(SCIE, IF 4.2)**
30. Jae-Hak Jeong, Bo-Mi Lee, Jun-Ki Hong, Tae-Heon Yang, **Yong-Hwa Park**, "Reproduction of human blood pressure waveform using physiology-based cardiovascular simulator," *Scientific Reports*, Vol. 13, No. 1, 2023.5; **(SCIE, IF 4.9) (Representative paper)**
31. Andersen Peter Risby, Gyeong-Tae Lee, Nielsen Daniel Gert, Kook Junghwan, Henriquez Vicente Cutanda, Aage Niels, **Yong-Hwa Park**, "Experimental characterization of a shape optimized acoustic lens: Application to compact speakerphone design," *Journal of The Acoustical Society of America*, Vol. 153, No. 4, 2351-2361, 2023.4; **(SCIE, IF 2.6)**
32. Won-Ho Jung, Seong-Hu Kim, Sung-Hyun Yun, Jae-Woon Bae, **Yong-Hwa Park**, "Vibration, acoustic, temperature, and motor current dataset of rotating machine under varying operating conditions for fault diagnosis," *Data in Brief*, Vol. 48, 2023.6; **(ESCI, IF 1.9)**
33. Wonho Jung, Sung-Hyun Yun, Yoon-Seop Lim, Sungjin Cheong and **Yong-Hwa Park**, "Vibration and Current Dataset of Three-Phase Permanent Magnet Synchronous Motors with Stator Faults," *Data in Brief*, Vol. 47, 108952, 2023.04**; (ESCI, IF 1.9)**
34. Gil -Yong Lee and **Yong-Hwa Park**, "A proper generalized decomposition-based harmonic balance method with arc-length continuation for nonlinear frequency response analysis," *Computers & Structures*, Vol. 275, 2023.01; **(SCIE, IF 5.6)**
35. Gyeong-Tae Lee, Hyeonuk Nam, Seong-Hu Kim, Sang-Min Choi, Youngkey Kim, **Yong-Hwa Park**, "Deep learning-based cough detection camera using enhanced features," *Expert Systems with Applications*, Vol.206, 2022.11; **(SCIE, IF 9.4) (Representative paper)**
36. Sung-Hyun Lee, Wook-Hyeon Kwon, Yoon-Seop Lim and **Yong-Hwa Park**, "Highly precise AMCW time-of-flight scanning sensor based on parallel-phase demodulation," *Measurement*, Vol.203, 2022.11; **(SCIE, IF 6.1) (Representative paper)**
37. Bomi Lee, Jae-Hak Jeong, Junki Hong and **Yong-Hwa Park**, "Correlation analysis of human upper arm

parameters to oscillometric signal in automatic blood pressure measurement," *Scientific Reports*, Vol.12, 2022.11; **(SCIE, IF 4.9)**

38. Gil -Yong Lee, K. C. Park and **Yong-Hwa Park**, "Reduced-order modeling via proper generalized decomposition for uncertainty quantification of frequency response functions," *Computer Methods in Applied Mechanics and Engineering*, Vol.401, 2022.11; **(SCIE, IF 7.6) (Representative paper)**

39. Jaewoong Bae, Wonho Jung, and **Yong-Hwa Park**, "Normal data based rotating machine anomaly detection using CNN with self-labeling," *Smart Structures and Systems*, Vol. 29, No.6, pp. 757-766, 2022; **(SCIE, IF 1.4)**

40. Gil-Yong Lee, **Yong-Hwa Park**, "A proper generalized decomposition based Padé approximant for stochastic frequency response analysis," *International Journal for Numerical Methods in Engineering*, Vol.122, No.22, pp.6596-6622, 2021; **(SCIE, IF 3.3)**

41. Daehee Park, **Yong-Hwa Park**, "Identifying Reflected Images from Object Detector in Indoor Environment Utilizing Depth Information," *IEEE Robotics and Automation Letters*, Vol. 6, No. 2, pp. 635-642, 2021.1; **(SCIE, IF 5.3)**

42. Yugang Chen, Kang-Jae Park, Bomi Lee, **Yong-Hwa Park**, "A method for identification of analyte concentration in aqueous solution by using a mistuned fluid-structure coupled vibration," *Journal of Mechanical Science and Technology*, Vol. 35, No. 2, pp.1-6, 2021.1; **(SCIE, IF 2.2)**

43. Gil-Yong Lee, Seung-Seop Jin, **Yong-Hwa Park**, "A natural frequency sensitivity-based stabilization in spectral stochastic finite element method for frequency response analysis*," Structural Engineering and Mechanics*, Vol. 75, No. 3, pp.311-325, 2020.8; **(SCIE, IF 3.2)**

44. Yugang Chen, **Yong-Hwa Park**, "A Helmholtz resonator on elastic foundation for measurement of the elastic coefficient of human skin," *Journal of The Mechanical Behavior of Biomedical Materials*, Vol. 101, pp. 1-7, 2020.1; **(SCIE, IF 4.1)**

45. Jeongjae Ryu, Hanbert Jeong, Yugang Chen, Chungik Oh, Jaegyu Kim, Hongjun Kim, Seongwoo Cho, Kwangsoo No, **Yong-Hwa Park**, Steve Park, Seungbum Hong, "Flexible Piezoelectric Liquid Volume Sensor," *Sensors and Actuators A: Physical*, Vol.276, pp.219~225, April 2018; **(SCIE, IF 5.1)**

46. Byung-Hoon Na, Gun-Wu Ju, Chang-Young Park, Soo-Kyung Lee, Hee-Ju Choi, Yong-Chul Cho, **Yong-Hwa Park**, Yong-Tak Lee, "Electro-Absorption Modulator with an Asymmetric Coupled Triple Quantum Well for Low-Voltage Operation," *Optics Letters*, Vol.40, No.14, pp.3376~3379, July 2015; **(SCIE, IF 3.3)**

47. Gun-Wu Ju, Byung-Hoon Na, **Yong-Hwa Park**, Young-Min Song, Yong-Tak Lee, "Recent Approaches for Broadening the Spectral Bandwidth in Resonant Cavity Optoelectronic Devices," *Advances in Condensed Matter Physics*, Vol.2015, pp.1~11, July 2015; **(SCIE, IF 3.1)**

48. Byung-Hoon Na, Gun-Wu Ju, Hee-Ju Choi, Soo-Kyung Lee, Sooraj Ravindran, Yong-Chul Cho, **Yong-Hwa Park**, Chang-Young Park, Yong-Tak Lee, "Coupled Tandem Cavities based Electro-Absorption Modulator with Asymmetric Tandem Quantum Well for High Modulation Performance at Low Driving Voltage," *Optics Express*, Vol.21, No.23, pp.27924~27932, November 2013; **(SCIE, IF 3.4)**

49. Sang-Hun Lee, Chang-Young Park, Jang-Woo You, Heesun Yoon, Yong-Chul Cho, **Yong-Hwa Park**, "850 nm IR Transmissive Electro-Absorption Modulator using GaAs Micromachining," *Sensors and Actuators A: Physical*, Vol.197, pp.47~52, August 2013; **(SCIE, IF 5.1)**

50. **Yong-Hwa Park**, Yong-Chul Cho, Jang-Woo You, Chang-Young Park, Hee-Sun Yoon, Sang-Hun Lee, Jong-Oh Kwon, Seung-Wan Lee, Byung-Hoon Na, Gun-Wu Ju, Hee-Ju Choi, Yong-Tak Lee, "Three-Dimensional Imaging using Fast Micromachined Electro-Absorptive Shutter," *Journal of Micro-nanolithography MEMS and MOEMS*, Vol.12, No.2, April 2013; **(SCIE, IF 2.0)**

51. Byung-Hoon Na, Gun-Wu Ju, Hee-Ju Choi, Yong-Chul Cho, **Yong-Hwa Park**, Chang-Young Park, Yong-Tak Lee, "Wide Spectral Bandwidth Electro-Absorption Modulator using Coupled Micro-Cavity with Asymmetric Tandem Quantum Well," *Optics Express*, Vol.20, No.17, pp.19511~19519, August 2012; **(SCIE, IF 3.4)**

52. Byung-Hoon Na, Gun-Wu Ju, Choi, Hee-Ju Choi, Yong-Chul Cho, **Yong-Hwa Park**, Yong-Tak Lee, "Large Aperture Asymmetric Fabry Perot Modulator based on Asymmetric Tandem Quantum Well for Low Voltage Operation," *Optics Express*, Vol.20, No.6, pp.6003~6009, March 2012; **(SCIE, IF 3.4)**

53. Hee-Moon Jeong, **Yong-Hwa Park**, Yong-Chul Cho, Junsik Hwang, Seok-Mo Chang, Seok-Jin Kang, Hyun-Ku Jeong, Jun O. Kim, Jin-Ho Lee, "Slow Scanning Electromagnetic Scanner for Laser Display," *Journal of Micro-nanolithography MEMS and MOEMS*, Vol.7, No.4, October 2008; **(SCIE, IF 2.0)**

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PROJECTS GRANTED (109 R&D projects with USD 20.62M grants total, at KAIST, 92 projects with USD 11.23M as PI)

01/2026-12/2026 Hankook Tire & Technology Co., Ltd., "Artificial Intelligence-Based Tire Integrity Monitoring Technology Development" (Research grant: USD 75.7K)

10/2025-09/2030 Samsung Electronics, “Development of a Multimodal Sensing-Based Predictive Fault Diagnosis System for Semiconductor Manufacturing Equipment” (Research grant: USD 266.7K)

09/2025-08/2026 LG Electronics Inc., “Development of a New Framework for Psychoacoustic Analysis and Level Quantification of Compressor Noise” (Research grant: USD 58.7K)

08/2025-12/2025 IITP, “Establishment of a Leading Physical AI Model and PoC” (Research grant: USD 43.1K)

07/2025-12/2025 HD KSOE, “Feasibility Study on the Development of an Autonomous Mobile System to Overcome Obstacles in Autonomous Mobility” (Research grant: USD 58.7K)

07/2025-12/2025 InnoCORE, “PRISM-AI: Platform for Real-world Integration of Smart Manufacturing and Artificial Intelligence” (Research grant: USD 43.3K)

05/2024-04/2026 LG Electronics Inc., “Development of a Refrigerator Installation-Space Scanning Kit” (Research grant: USD 73.3K)

05/2025-09/2025 Samsung Heavy Industry Co., “Development of a Prediction Method Based on Multivariate Time-Series Data” (Research grant: USD 22.0K)

03/2025-02/2029 Research Institute of Medium & Small Shipbuilding, “Super-Gap Shipbuilding Industry Expert Training Program” (Research grant: USD 1.46M)

03/2025-11/2025 HD Hyundai Electric, “Development of a Vibration Signal-Based Diagnostic Algorithm for C-GIS Operating Mechanisms” (Research grant: USD 88.0K)

01/2025-01/2027 POSCO, “Development of a VR-Based Real-Time Fault Diagnostic Device Using Motion Amplification” (Research grant: USD 151.3K)

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05/2024-04/2029 National Research Foundation of Korea, “Development of Non-invasive Cardiovascular Health Monitoring Technology Using Virtual Patient System and Physics-informed AI” (Research grant: USD 0.8M)

04/2024-06/2024 KIST, “Implementation of a Deep Learning-Based Model for Processing Auditory Perception EEG Data” (Research grant: USD 3.0K)

01/2024-12/2024 KRISO, “A Study on the Detection of Sound Events Robust in the Noise Environment in Ships (2024)” (Research Grant: USD 74.0K)

01/2024-05/2024 Agency for Defense Development, "Research on Diagnosing the Powertrain of K1 Tanks with an Artificial Intelligence Fault Diagnosis System based on Acoustic-Vibration Data" (Research grant: USD 0.093M)

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11/2023-02/2024 Samsung Heavy Industry Co., Ltd., “Application and verification of solid line for machine independent rotating body failure prediction diagnosis (2023)” (Research Grant: 74K)

11/2023-10/2026 Agency for Defense Development Institute of Civil-Military Technology Cooperation, "(RCMS)XR-Based Minefield Formation Training System Development" (Research grant: USD 0.416M)

07/2023-12/2022 KAIST, "Environmentally Adaptive Vibration/Current-Based Motor AI Fault Diagnosis and Life Prediction (2023)" (Research grant: USD 11.4K)

07/2023-01/2024 LG Electronics Inc., "Sound Camera Post-Processing Software AI Algorithm Development (2023)" (Research grant: USD 68.1K)

07/2023-06/2024 Hankook Tire & Technology Co., Ltd., "Artificial Intelligence-Based Tire Integrity Monitoring Technology Development (2023)" (Research grant: USD 75.7K)

06/2023-05/2024 KEPCO Research Institute, "Development of Wind Turbine Tower Structural Health Monitoring Technique based on Artificial Neural Network and Computational Dynamics (2023)" (Research grant: USD 37.8K)

06/2023-11/2023 KITECH, "Defect WGB Dynamic Response Data Measurement and Analysis Service (2023)" (Research grant: USD 26.5K)

05/2023-07/2023 KITECH, "WGB Fatigue Damage Threshold Measurement and Analysis (2023)" (Research grant: USD 12.1K)

05/2023-09/2023 Samsung Heavy Industry Co., Ltd., "Development of Machine-Independent Artificial Intelligence-Based Anomaly Detection Technique (2023)" (Research grant: USD 30.3K)

03/2021-02/2026 Research Institute of Medium & Small Shipbuilding, "(RCMS) Smart Yard Professional Workforce Development (2023)" (Research grant: USD 1.89M)

01/2023-12/2023 KRISO Korea Research Institute of Ships & Ocean Engineering, "Development of Acoustic Recognition AI Model for Situational Awareness and Anomaly Detection within Ships (2023)" (Research grant: USD 0.114M)

01/2023-12/2023 KAIST, "Development of Structural Health Monitoring Technique for HVAC Facilities Based on Digital Twin and Artificial Neural Network (2023)" (Research grant: USD 45.4K)

01/2023-12/2023 Agency for Defense Development Institute of Civil-Military Technology Cooperation, "(RCMS) Research on Diagnosing the Powertrain of K1 Tanks with an Artificial Intelligence Fault Diagnosis System based on Acoustic-Vibration Data (2023)" (Research grant: USD 0.136M)

10/2022–12/2023 Korea Shipbuilding & Offshore Engineering, "Development of High-Speed/High-Precision Measurement Technology for Hull Block Alignment" (Research grant: USD 0.091M)

11/2022-08/2023 LG Electronics Inc., "Development of Compressor Journal Bearing Wear Diagnosis Technology Using Current Signal Imaging Technique" (Research grant: USD 60.5K)

10/2022-12/2022 KITECH, "Analysis of Fault Characteristics in Harmonic Reducers" (Research grant: USD 14.4K)

10/2022-11/2022 Samsung Heavy Industry Co., Ltd., "Fault Diagnosis of Rotating Machinery Using Image Data in Adverse Conditions" (Research grant: USD 7.57K)

08/2022-11/2022 Samsung Heavy Industry Co., Ltd., "Feasibility Study on Early Detection of Arc Discharge in Yard/Ship Switchboards based on Acoustic Recognition Technology" (Research grant: USD 7.57K)

08/2022-07/2024 Korea East-West Power Co., Ltd, "Development of a Smart Safety Diagnosis Platform for Switchboards using Artificial Intelligence-based Acoustic Recognition Technology" (Research grant: USD 0.144M)

07/2022-12/2022 KAIST, "Development of AI Fault Diagnosis for Production Facility Motors based on Current and Normal Data" (Research grant: USD 11.4K)

07/2022-09/2022 Samsung Heavy Industry Co., Ltd., "Motor Fault Diagnosis Using Multivariate Time Series of Current Data in Adverse Conditions" (Research grant: USD 7.57K)

06/2022-12/2023 Agency for Defense Development, "Research on Diagnosing the Powertrain of K1 Tanks with an Artificial Intelligence Fault Diagnosis System based on Acoustic-Vibration Data" (Research

grant: USD 0.473M)

06/2022-05/2023 KEPCO Research Institute, "Development of Structural Health Monitoring Technique for Wind Turbine Tower based on Artificial Neural Network and Computational Dynamics" (Research grant: USD 37.8K)

05/2022-06/2022 Samsung Heavy Industry Co., Ltd., "Predicting the Service Life of Rotating Machinery using Multivariate Time Series of Vibration Normal Data in Adverse Conditions" (Research grant: USD 7.57K)

04/2022-03/2023 LS Cable & System, "Development of Fault Diagnosis Technology for Drive Motors based on Normal Data" (Research grant: USD 0.136M)

03/2022-12/2023 KAIST, "Development of Structural Health Monitoring Technique for HVAC Systems based on Digital Twin and Artificial Neural Network" (Research grant: USD 0.0984M)

03/2022-04/2022 Samsung Heavy Industry Co., Ltd., "Fault Diagnosis of Rotating Machinery using Multivariate Time Series of Vibration Normal Data in Adverse Conditions" (Research grant: USD 7.57K)

01/2022-06/2022 KAIST, “Numericalization of Data Augmentation Distortion for Optimization of Deep Learning-Based Acoustic Event Detection (2022)” (Research grant: USD 1.5K)

11/2021-02/2022 Samsung Heavy Industry Co., Ltd., "Feasibility Verification of Improved Motion Amplification Algorithm using Camera Vibration Characteristics" (Research grant: USD 7.57K)

11/2021-06/2021 KAIST, "Data Collection for Permanent Magnet Motor Fault Diagnosis" (Research grant: USD 37.8K)

07/2021-12/2021 KAIST, "Development of Motion Amplification Technology and Digital Twin Modeling Technique for Vibration Signal-Based Anomaly Diagnosis in Manufacturing Facilities" (Research grant: USD 15.1K)

01/2022-12/2022 KIOST, "Development of AI Models for Object Recognition and Navigation Equipment Information (Visual, Acoustic) Recognition for Onboard Situation Awareness" (Research grant: USD 0.027M)

03/2021-12/2021 KIOST, "Development of AI Models for Object Recognition and Navigation Equipment Information (Visual, Acoustic) Recognition for Onboard Situation Awareness" (Research grant: USD 0.024M)

12/2020–12/2021 Korea Shipbuilding & Offshore Engineering, "Development of High-Speed/High-Precision Measurement Technology for Hull Block Alignment" (Research grant: USD 0.060M)

10/2020–12/2022 KEPCO Nuclear Fuel, "Development and Verification of Control Rod Drop Simulation Technology Considering Seismic Conditions" (Research grant: USD 0.28M)

04/2020–07/2020 Samsung Heavy Industry Co., Ltd, "Discovery of Background Vibration Removal Algorithm for the Application of Video Diagnostic System Inside Ships" (Research grant: USD 7.57K)

01/2022–12/2022 Ministry of Oceans and Fisheries, "Development of a Korean All-Weather Rescue Boat for Coastal Isolated Individuals (Subproject: Development of Image Algorithm for Rescuer Recognition)" (Research grant: USD 0.036M)

01/2021–12/2021 Ministry of Oceans and Fisheries, "Development of a Korean All-Weather Rescue Boat for Coastal Isolated Individuals (Subproject: Development of Image Algorithm for Rescuer Recognition)" (Research grant: USD 0.053M)

04/2020–12/2020 Ministry of Oceans and Fisheries, "Development of a Korean All-Weather Rescue Boat for Coastal Isolated Individuals (Subproject: Development of Image Algorithm for Rescuer Recognition)" (Research grant: USD 0.059M)

08/2019–03/2020 Ministry of Oceans and Fisheries, "Development of a Korean All-Weather Rescue Boat for Coastal Isolated Individuals (Subproject: Development of Image Algorithm for Rescuer Recognition)" (Research grant: USD 0.044M)

04/2020–11/2021 KEPCO Nuclear Fuel, "Development and Validation of Seismic Analysis Evaluation System for Nuclear Fuel" (Research grant: USD 0.0908M)

02/2020–01/2023 InBody, "Correlation Analysis of Oscillometric Signals, K-Sound Signals, and Cardiovascular Parameters" (Research grant: USD 0.227M)

04/2020–07/2020 KAIST, "Development of Sensor Network and AI-Based Situation Awareness Technology for Disaster Rescue" (Research grant: USD 37.8K)

09/2019-12/2020 Korea Shipbuilding & Offshore Engineering Co., Ltd, “Feasibility Study on the Development of High-Speed/High-Precision Measurement Technology for Hull Blocks” (Research grant: USD 151.4K)

06/2019–05/2022 Intekplus Co., Ltd., "Development of Miniaturized 3D Sensor Platform" (Research grant: USD 0.227M)

01/2021–12/2021 Ministry of Science and ICT, “Development of 3D scene recognition module and algorithm for silver care robots. (Total Research grant: USD 0.030M)

01/2020–12/2020 Ministry of Science and ICT, “Development of 3D scene recognition module and algorithm for silver care robots. (Total Research grant: USD 0.027M)

01/2019–12/2019 Ministry of Science and ICT, “Development of 3D scene recognition module and algorithm for silver care robots. (Total Research grant: USD 0.018M)

01/2018–12/2018 Ministry of Science and ICT, “Development of 3D scene recognition module and algorithm for silver care robots. (Total Research grant: USD 0.014M)

12/2019–08/2020 Korea Shipbuilding & Offshore Engineering, "Development of Real-time Absence Shape Measurement and Automatic Extraction Technology of Corner Trajectory" (Research grant: USD 0.182M)

12/2020–05/2021 KAERI, “Modal test of small-sized model for reactor assembly and validation of modeling method” Vibration analysis of new reactor system and development of finite element model. (Research grant: USD 0.021M)

12/2019–11/2020 KAERI, “Modal test of small-sized model for reactor assembly and validation of modeling method” Vibration analysis of new reactor system and development of finite element model. (Research grant: USD 0.070M)

12/2018–11/2019 KAERI, “Modal test of small-sized model for reactor assembly and validation of modeling method” Vibration analysis of new reactor system and development of finite element model. (Research grant: USD 0.17M)

01/2020–12/2020 Korea Institute of Machinery and Materials (KIMM), “Development of 3D Scene Recognition Solution for Autonomous Working Robot” Development of a compact 3D imagining device and its signal processing scheme for obstacle detection and 3D mapping of autonomous working robot systems in small to medium size factories. (Research grant: USD 0.63M)

01/2019–12/2019 Korea Institute of Machinery and Materials (KIMM), “Development of 3D Scene Recognition Solution for Autonomous Working Robot” Development of a compact 3D imagining device and its signal processing scheme for obstacle detection and 3D mapping of autonomous working robot systems in small to medium size factories. (Research grant: USD 0.074M)

01/2018–12/2018 Korea Institute of Machinery and Materials (KIMM), “Development of 3D Scene Recognition Solution for Autonomous Working Robot” Development of a compact 3D imagining device and its signal processing scheme for obstacle detection and 3D mapping of autonomous working robot systems in small to medium size factories. (Research grant: USD 0.074M)

01/2019–12/2019 SUNG SHIN ROLLING STOCK TECHNOLOGY Ltd., “Development of rail flow detecting car for the monitoring system of rail damage inspection and concrete crack detection module” Experimental vibration analysis and finite element model of new rail investigation car (Research grant: USD 0.060M)

01/2018–12/2018 SUNG SHIN ROLLING STOCK TECHNOLOGY Ltd., “Development of rail flow detecting car for the monitoring system of rail damage inspection and concrete crack detection module” Experimental vibration analysis and finite element model of new rail investigation car (Research grant: USD 0.058M)

09/2017–12/2017 SUNG SHIN ROLLING STOCK TECHNOLOGY Ltd., “Development of rail flow detecting car for the monitoring system of rail damage inspection and concrete crack detection module” Experimental vibration analysis and finite element model of new rail investigation car (Research grant: USD 0.022M)

03/2019–2/2020 National Research Foundation (NRF), “Development of 3D Image Sensor for Autonomous Vehicles” Development of a compact and low-cost 3D sensor for autonomous vehicles. (Research grant: USD 0.10M)

03/2018–2/2019 National Research Foundation (NRF), “Development of 3D Image Sensor for Autonomous Vehicles” Development of a compact and low-cost 3D sensor for autonomous vehicles. (Research grant: USD 0.10M)

03/2017–2/2018 National Research Foundation (NRF), “Development of 3D Image Sensor for Autonomous Vehicles” Development of a compact and low-cost 3D sensor for autonomous vehicles. (Research grant: USD 0.16M)

01/2019–12/2019 Long-term URP (Undergraduate Research Program), KAIST, “Basic study for the biometric authentication utilizing vibration characteristics of human fingers” Basic study on the human finger vibration characteristics for biometric application of future mobile devices (Research grant: USD 3.03K)

01/2019–12/2019 Long-term URP (Undergraduate Research Program), KAIST, “Development of Smart Active Sheet for Deep Sleep” Basic study and prototyping of comport and active bed system for deep sleep based on the sleep science. (Research grant: USD 3.78K)

03/2018–10/2018 RED&B, KAIST, “Development and project planning of a core technology used for intelligent robot capable of automatic operation in the small and medium sized factories” Project planning with KIMM engineers for autonomous robot development (Research grant: USD 30.3K)

03/2018–10/2018 URP (Undergraduate Research Program), KAIST, “3D depth method using optical flow” Preliminary study and experiment for new 3D imaging device concept. (Research grant: USD 1.51K)

06/2017–12/2017 URP (Undergraduate Research Program), KAIST, “Pre-research for Development of Non-invasive Blood Glucose Level Measurement: Machine Learning Algorithm for Blood Glucose Level and Human Dynamic Modulus” Preliminary study and experiment for new non-invasive glucose sensing device concept based on the vibration signal of human body. (Research grant: USD 1.51K)

06/2017–12/2017 K-School KAIST, “Development of Core Technology of Non-invasive, Portable Diabetes Measurement Device” Group research for new non-invasive glucose sensing device concept based on the vibration signal of human body. (Research grant: USD 0.151M)

01/2017–06/2017 URP (Undergraduate Research Program), KAIST, “Simulation study and verification of distortion-decreased imaging for a robust three-dimensional image sensor” Enhancement of 3D depth algorithm using signal processing considering nonlinearities (Research grant: USD 1.51K)

12/2016–02/2017 K-School, KAIST, “Development of Non-invasive, Portable Diabetes Measurement Device” Research plan of group research for new non-invasive glucose sensing device concept based on the vibration signal of human body (Research grant: USD 15.1K)

08/2016–12/2019 Department of Mechanical Engineering, KAIST, “Signal processing of 3D image sensor and search for new compact module for autonomous driving applications” Development of a novel 3D imagining device and its signal processing scheme for obstacle detection and 3D mapping of autonomous robot systems and vehicle applications. (Research grant: USD 0.151M)

<u>Projects at Samsung Electronics</u>

03/2013–06/2016 Device and System Research Center, Samsung Advanced Institute of Technology, Samsung Electronics, “Development of 3D Image Sensor for User Recognition in Display System” and “Development of 3D Image Sensor for Autonomous Robot Navigation” Development of next generation-3D imagining device for obstacle detection and indoor 3D mapping of autonomous

robot systems for future home applications. (Research grant: USD 4.62M)

11/2010–02/2013 Device and System Research Center, Samsung Advanced Institute of Technology, Samsung Electronics, “Development of 3D Image Sensor for User Recognition in Display System” Development of next generation-3D imagining system for user interface and gesture recognition for IT device applications. (Research grant: USD 1.89M)

03/2013-02/2014 Device and System Research Center, Samsung Advanced Institute of Technology, Samsung Electronics, "Development of Active Sensor-Based HD-Grade 3D Depth Camera" Industrial Convergence Frontier Technology Development Project, National Project by the Ministry of Knowledge Economy. Participating Organization: KETI, Joint Collaboration with Kwangwoon University (Participating Research, Total Research grant: USD 2.88M)

01/2008–10/2010 Micro Systems Lab, Samsung Advanced Institute of Technology, Samsung Electronics, “Development of 3D Camera” Development of next generation-3D imagining system for consumer camera devices. (Research grant: USD 1.36M)1/2007–1/2008 Micro Systems Lab, Samsung Advanced Institute of Technology, Samsung Electronics, “Design for Reliability of Scanning Display Device” Development of ultra-small mobile display systems. MEMS resonating mirror design and optimization.

3/2005–1/2007 Micro Systems Lab, Samsung Advanced Institute of Technology, Samsung Electronics, “Development of 2D MEMS scanner” Development of MEMS scanning display system. 2D resonating MEMS scanner design and optimization.

8/2004 –2/2005 Visual Display Division, Samsung Electronics, “Development of LED based DLP small projector” Conducted joint development with Carl Zeiss, Osram and TI (International Consortium).

4/2004 –12/2004 Visual Display Division, Samsung Electronics. Global collaboration with Russian Society of Academy. Initiated Display Technology LAB in Samsung Research Center (SRC) in Moscow. Conduct three oversea outsourcing projects in MEMS display area with Bauman State University in Moscow.

2/2004 –2/2005 Visual Display Division, Samsung Electronics, “Development of LED based Digital Light Processing (DLP) Rear Projection TV” Conducted global joint development of next generation DMD package systems in collaboration with Samsung semiconductor division and Texas Instrument. Development of next generation DLP systems (HD3 diamond pixel DMD and its optical module).

Projects at University of Colorado at Boulder

1/2002 –6/2003 Univ. of Colorado at Boulder, “Optimization of Switching Speed and Actuation Energy of RF MEMS Switches” sponsored by Raytheon. Conduct electro-thermal-fluid-structural coupled analysis of MEMS switch. Design optimization for high-speed, low-loss MEMS switches for RF circuit applications.

10/2000 –6/2003 Univ. of Colorado at Boulder, “Dynamics of MEMS resonator for loss modeling and design” sponsored by Coventor Ware Inc. Electro-mechanical coupled modeling for MEMS resonator and filter. Identified energy loss mechanism and improved dynamics characteristics of RF micro resonators by multi-physics micro-system simulation.

9/2000 –6/2003 Univ. of Colorado at Boulder, “Modeling, Simulation and Validation of High-Fidelity Structural Dynamical Systems: Partitioned Structural Eigenvalue Analysis” sponsored by NSF-SNL Grant CTS-9732179. Implementation of parallel computation of structural eigenvalue analysis for large-scale and massive computation.

Projects at KRISS and KAIST (RA)

9/1999-8/2000 Acoustics and Vibration Lab, KRISS, “Development of the Human Sensibility Evaluation Simulator” sponsored by the Ministry of Science and Technology. Conducted vibration transmission analysis of human body. Performed biodynamic modeling and identification.

9/1999-1/2000	NOVIC, KAIST, “Comparison Study on the Vibration Characteristics of Commercial Engine Blocks and Power-trains” sponsored by Hyundai Motor Company. Performed experimental modal analysis. Conducted FE Model updating and correlation analysis.
9/1998-8/1999	NOVIC, KAIST, “A Study on the Unified Analysis and Design Technique for Low-noise Vibro-acoustic Systems” sponsored by the Korea Science and Engineering Foundation (KOSEF). Conducted structural-acoustic coupling analysis of vehicle interiors. Design optimization.
9/1995-8/1998	NOVIC, KAIST, “Inverse Problem Solvers in Noise and Vibration Engineering: Methods and Applications” sponsored by KOSEF. Conducted indirect force identification, regularization of signatures, and placements of vibration measuring sensors.
9/1992-8/1995	NOVIC, KAIST, “Development of Basic Technologies for Low Noise/Vibration Vehicle Design” sponsored by KOSEF. Conducted FE and experimental modal analysis, FE model updating for white body vehicle. Semi-structural acoustic coupling analysis. Accomplished experimental structural optimization.
4/1993-3/1995	NOVIC, KAIST, “Acceleration Canceling Hydrophone Model Analysis” sponsored by ADD. Developed FE code for modal analysis of piezoceramic-polymer structure with MATLAB. Coupled field analysis of piezoceramic sensor.

CONSULTING (76)

2025.10.16	POSCO: Recent Technologies in Multimodal Condition Diagnosis Based on Vibration, Acoustics, and Vision (POSCO Group Chair Professor Forum)
2025.07.04	HD KSOE: Technical Exchange on Design and Analysis of Shipbuilding Structures Using Vibration Analysis and Generative AI
2025.06.30	SOLUM: Seminar and Technical Exchange on the Development of a Device for Predicting Cardiovascular Health Status Using Earphones and PPG
2025.04.30	Samsung Electronics: Samsung Electronics' Semiconductor Packaging Line On-Site Robotic Acoustic-Vibration Fault Diagnosis
2025.04.16~06.20	Bukang Tech: Consultation on the Design of a Data-Center Cooling System Using Wastewater Treatment (Served as KAIST ILP Consulting PD)
2025.02.26	LG Electronics: Development of Core Technologies for LG Electronics Compressors (UFRC Second-Year Discussion)
2024.09.05~06	Mirae Company: Consultation on Deriving a Mathematical Dynamics Model for Predicting Vibration of Mirae Company's Export Robot
2024.07.02	Iljin Group: Consultation on NVH Testing of Cylindrical Roller Bearings (Iljin Group BearingArt)
2024.04.10	Samsung Heavy Industries: Development and Practice of Smart Condition Monitoring and Diagnostic Technologies (Advanced Seminar on Smart Ship and Autonomous Navigation Technology Development)
2024.03.20	Hyundai Motor Company: Driver Health Condition Diagnosis Using Biometric Information (Focused on Automotive New Biz Models) (Hyundai Motor Convergence Technology Expert Technical Advisory Seminar)
2024.02.21	Sejong Science High School: Energy Harvesting Using Acoustic Diodes (R&E Mentoring)
2024.02.07	Regulatory Innovation Promotion Team, Office for Government Policy Coordination: Consultation on Auditory Analysis of Vehicle Horns and Minimization Measures (Expert Committee Member Han Seok-hoe)
2024.01.22	ROK Navy Submarine Force Command, Gyeryongdae: Development of a Visualization AI Platform Integrating Sonar Information (Sea-through 3D VR Technology)
2024.02.07	Samsung Heavy Industries: Ship Fault Diagnosis Technology: AI-Based Rotating Machinery Fault Diagnosis Using Vibration, Acoustic, and Image Data (SHI–KAIST Tech Forum)
2024.01.22	Winicare: Consultation on Sensors and Signal Processing for Motor Fault Diagnosis, AI Technology
2023.11.28	Doosan Enerbility: Turbine Dynamic Characteristic Variable Clarification and Fault Diagnosis Technology Consultation
2023.11.10	LG Electronics: Introduction and Consultation on Non-Invasive Blood Glucose Technology
2023.11.09	Equinor: High Fidelity Digital Twin for Health Monitoring of Maritime Infrastructures
2023.10.18	Hyundai: Development of a Smart Yard Crane Load Recognition System
2023.05.12	Hyundai Electric: A study on the motor fault diagnosis, consulting
2023.03.03	Silla Innotech: Business Consultation on Mattress Disassembly Automation Robot Technology, ILP Professor Advisory Panel2023.05.12 HD Hyundai Electric Co., Ltd.: A study on the motor fault diagnosis
2023.02.13	LIG Nex1: AI-Based Abnormal Diagnosis and Applications Using Vibration, Acoustic, and Image Data
2023.02.08	Cheongsong-gun, Gyeongsangbuk-do: Consultation for Cheongsong County Livestock Farming Project
2022.11.28	Hankook Tire & Technology Co., Ltd.: Smart Tire Fault Diagnosis Technology

	Exchange and Seminar
2022.10.07	Hyundai-Rotem Co.: Technology Exchange for Tank Fault Diagnosis
2022.10.19	POSCO CO., LTD.: POSCO Distinguished Forum Presentation (Advanced 3D Vision and Acoustic Sensors for Manufacturing Intelligence and Safety)
2022.09.18	Doosan Enerbility Co., Ltd.: Doosan Enerbility-KAIST Technology Exchange, Demo
2022.05.19	Samsung Heavy Industry Co., Ltd.: SHI-KAIST 28th Year Industry-Academia Cooperation Council
2022.03.23	Sejong Academy of Science and Arts: Graduation Research Advisory
2021.04.29	Samsung Heavy Industry Co., Ltd.: SHI-KAIST 27th Year Industry-Academia Cooperation Council
2022.10~2022.11	Samsung Heavy Industry (Daedeok): Fault Diagnosis of Rotating Machinery Using Image Data under Harsh Conditions
2022.08~2022.11	Samsung Heavy Industry (Pangyo): Feasibility Study on Early Detection of Arc Discharge in Yard/Ship Switchboards based on Acoustic Recognition Technology
2022.07~2022.09	Samsung Heavy Industry (Daedeok): Motor Fault Diagnosis Using Multivariate Time Series Normal Data of Current under Bad Conditions
2022.05~2022.06	Samsung Heavy Industry (Daedeok): Lifetime Prediction of Rotating Machinery using Multivariate Time Series Normal Data in Bad Conditions
2022.03~2022.04	Samsung Heavy Industry (Daedeok): Fault Diagnosis of Rotating Machinery using Multivariate Time Series Normal Data of Vibration under Bad Conditions
2021.11.19~2022.02.18	Samsung Heavy Industry (Pangyo): Validation of the Feasibility of Improved Motion Amplification Algorithm using Camera Vibration Characteristics
2022.08.30	LG Electronics Inc: LG-KAIST 2nd Half Industry-Academia Technology Exchange Meeting, Presentation on Compressor Fault Diagnosis
2022.03.31	LG Electronics Inc: LG-KAIST 1st Half Industry-Academia Technology Exchange Meeting, Presentation on Motor Fault Diagnosis
2021.03.25	LG-KAIST H&A Research Planning Project Operating Committee the 1st Session
2021.04.23	LG-KAIST H&A Research Planning Project Operating Committee the 2nd Session
2021.06.17	LG-KAIST H&A Research Planning Project Operating Committee the 3rd Session
2019.11.29~present	THK (Japan): Industry-Academia Collaboration, Discussion on Student Intern Exchange
2022.02.04	Jabra: Technology Collaboration Meeting and Exchange (Audio Recognition)
2021.05.31	Korean Air R&D Center: Technology Exchange and Discussion in Fault Diagnosis and Workforce Development Fields
2022.01.19~ present	SN Co., Ltd.: ILP Professor Advisory Team Corporate Advisory (Electro-Mechanical Drone Launcher)
2022.02.22~ present	ROTAREXLUZTECH Co., Ltd: ILP Professor Advisory Team Corporate Advisory (Semiconductor Valve Design)
2022.03.02~ present	Seondo Electric Co., Ltd: South-East Power R&D Incubating Project Corporate Advisory (Fault Diagnosis of Switchboards Using Acoustics)
2022.03.18~2022.04.20	Windrorobtics: South-East Power R&D Incubating Project Corporate Advisory (Robot Development for Fault Diagnosis in Wind Turbines)
2021.06.28~2021.08.27	BN STEELA Industries Co., Ltd: Data Infrastructure Establishment AI Consulting and AI Solution Pilot Support (Coating Process Tolerance Control)
2021.10.12	L Fin Co., Ltd: Biometric Technology Advisory
2021.10.14	JBridge Marketing Company: Advisory on Gas Detection Using Thermal Cameras
2022.09.14	DRGKOREA Co., Ltd: KAIST-Trade Association Technical Barrier Consultation Meeting (Advisory on Miniaturization of Golf Range Recognition Device)
2021.08.26	Samsung Heavy Industry Technology Development Division Seminar
2021.9.24	ARE: Introduction to Audio Recognition Technology
2021.11.29	Ministry of National Defense, Eulji Research Center: Future Defense Strategy

	Innovation Program
2021.11.10	Four Major Science Institutes Joint: Small and Medium-sized Enterprise R&D Shared Center Technical Presentation 2nd
2021.09.15	Four Major Science Institutes Joint: Small and Medium-sized Enterprise R&D Shared Center Technical Presentation 1st
2021.07.31	KIIT: Vibration Acoustic-Based Fault Diagnosis Seminar
2021.09.14	Samsung Heavy Industry Co., Ltd. (Ship and Ocean Research Institute): Acoustic Vibration-Based Fault Diagnosis Technology Seminar
2020.11.27	Korea Astronomy and Space Science Institute: Lunar Probe Light Field Camera Design Review
2020.11.6 ~ 2021.1.5	SUNG-O PRECISIONS: Data Infrastructure Establishment AI Consulting and AI Solution Pilot Support
2020.10.8~ 2020.12.7	Donga Engineering Company: Data Infrastructure Establishment, AI Consulting, and AI Solution Proof-of-Concept Support
2020.9.5~ 2021.1.31.	UNBARRIER, INC.: Sign Language Interpretation Wearable Device Signal Processing and Sensor Advisory
2020.9.4. ~ 2020. 12.31	SN Co., Ltd, Korea South-East Power Co.: Fault Diagnosis Using EES Acoustics
2020.9.1	YTN: Infrared Camera Temperature Measurement Consultation
2020.06.01-2021.02.28	Education Institute for Gifted/ Incheon Science High School: Expert Advisory for Science Gifted Creative Research (R&E) Support Center
2020.4.16~ 2020.7.15	Samsung Heavy Industry (Pangyo): Background Vibration Removal Algorithm for Shipboard Video Diagnostic Systems
2019.12.14	Samsung Heavy Industry: SHI-KAIST Industry-Academia Cooperation Council
2019.11.30	SKYSYS CO., Ltd: Resolution of Vibration Issues in Images
2019.10.1~ 2020.1.31	Korea Shipbuilding & Offshore Engineering Co., Ltd: Innovation Technology Discovery for New Growth Engines
2019.9.1~ 2019.12.31	Korea Shipbuilding & Offshore Engineering Co., Ltd: Feasibility Study on the Development of High-Speed/High-Precision Measurement Technology for Hull Blocks
2019.7.18~19	Samsung Heavy Industry (GEOJE): Future Challenges Workshop and Technical Advisory
2019.06.28	Samsung Heavy Industry (GEOJE): Introduction to Future Technologies
2019.05.10	Samsung Heavy Industry: SHI-KAIST Industry-Academia Cooperation Council
2019.4.16	LG Innotek Co., Ltd.: Camera Technology External Expert Interview
2019.3.27	HD Hyundai Heavy Industries: Resolution of Vibration Issues in Welding Workshops
2018.7.11.~7.13	Chrome: Venture Enterprise Technology Advisory 2nd Session, Human Vibration Measurement Sensor Advisory
2018.6.18	INTEKPLUS CO., LTD.: Expert Advisory on the Latest Technology in the 3D Sensing

FELLOWSHIPS

Research Master	Samsung Advanced Institute of Technology, Samsung Electronics. 2/2012-7/2016
International Expert	TRIZ Expert Level 2, International TRIZ Association, 2008-present
Fellowship	Oversea post-doctoral fellowship of Korea Sci. and Eng. Foundation (KOSEF) 9/2000-8/2001.
Fellowship	Support program for future researchers of Korea Research Foundation (KRF), 9/1998-8/1999.
Lifetime Member	The International Society for Optical Engineering (SPIE) since 2012
Lifetime Member	Korean Society of Mechanical Engineers (KSME) since 2016
Lifetime Member	Korean Society for Noise and Vibration Engineering (KSNVE) since 2017
Member	IEEE, Acoustical Society of Korea (ASK) since 2017

EXTRACURRICULAR ACTIVITIES

Supervisor	KAIST Micro Robot (MR) Research Association (2017-present)
Supervisor	KAIST Research Associate for Creationism and Science (RACS) (2017-present)
Chairman	TRIZ Association of Samsung Advanced Institute of Technology, 2010-2016
Chairman	Graduate student's organization of NOVIC, KAIST (1995)
Chairman	Student session of the first Korea-Japan Symposium of Frontiers in Vibration Science and Technology (9/1998)
Championship	KAIST Open Tennis Tournament, Double match. (5/2000)
Championship	KAIST Open Tennis Tournament, Single match. (10/1996)
Voluntary Service	Served as a phone counselor in Life Line at KAIST (3/1997-8/1999)
Chairman	Research Associate for Creationism and Science (RACS, 6/1994-8/1999)
Executive member	Graduate Students Organization of KAIST (3/1991-8/1991)
Chairman	KAIST Mechanical Engineering Student Association (1988-89)
Chairman	KAIST Micro Robot (MR) Research Association (3/1989-2/1990)

MEDIA·PRESS

2026.4	"KAIST Undergraduates Advance to the Finals of the World's Largest Mars Exploration Competition" (The Korea Economic Daily, ETNews, Daylian, etc).
2025.8	"Expanding Field Experience through Industry-University Cooperative (Co-op) Programs (the Hankook Ilbo)
2023.5	"The Eyes of the Fourth Industrial Revolution: LiDAR (Light Detection and Ranging)" - A scientific perspective on the world, SEE Episode 104, KBS 1TV.
2023.3	"Mega Trends in Autonomous Driving Sensors and Digital Health Trends Through CES 2023," DT Magazine (KOITA)
2023.1.6	"Insights into CES 2023 Trends by Top Professors from KAIST," KAIST-Korea Economic Daily CES 2023 Special Joint Reporting Team, Online Webinar (YouTube Broadcast) and Newspaper Promotion
2020.8.3	"Deep Learning-based Cough Detection Camera" News broadcasting on KBS, MBC, JTBC, YTN; More than 70 news articles in Yonhap News, U.K. Daily Mail, and YouTube.

End of C.V.